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- Video: Public-Key Certificate and Public-Key Infrastructure (PKI) 5 min
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• Congratulations! You passed!

Grade received 86.90%

Latest Submission Grade 85.71%

Key Distribution and Management

Review Learning Objectives

1. Alice and Bob wants to communicate with each other. They first use Diffie-Hellman Key Exchange protocol to establish a session key and then use the session key to encrypt the messages from Alice to Bob. An attacker eavesdrops on the resulting ciphertext and conducts a brute-force attack on the encryption. Suppose the attacker succeeds in her brute-force search. Which of the following is NOT a purpose of a nonce in key distribution? Select all that applies. 3 / 3 points

Due May 5, 11:59 PM IST Attempts 3 every 8 hours

Try again

Retake the quiz in 7h 57m

- ☐ Bob's public key
- ☒ Bob's private key
- ☒ Correct See Key Hierarchy lecture.

Receive grade To Pass 75% or higher

Your grade 86.90%

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- ☐ The message from Alice to Bob
- ☐ The session key
- ☒ Alice's private key
- ☒ Correct See Key Hierarchy lecture.

2. Which of the followings is the most temporary by design? 3 / 3 points

- ☐ Master key
- ☐ Public-key certificate
- ☒ Nonce
- ☐ A user's public-private key pair
- ☐ Session key
- ☒ Correct See Key Hierarchy lecture.

3. What are the purposes of a nonce in key distribution? Select all that applies. 0 / 3 points

- ☒ To use it as a replacement key in case of compromise
- ☒ This should not be selected
- ☐ It is used by convention and has no practical use
- ☐ To use it for encryption so that the ciphertext does not make sense to the attacker
- ☐ To connect the response to the corresponding communication
- ☐ For padding the packet

4. There are 5 computers (hardware). Each computer hosts 4 virtual machines (VM) with different addresses, and each VM hosts 10 applications. How many key exchange sessions (in one session, two logical parties share a key) are needed if the decentralized key distribution is used so that all VM's share a pair-wise keys? 3 / 3 points

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☒ Correct See Key Hierarchy and Decentralized Key Distribution lecture.

5. ([This problem has the same setup as the previous question but asks a different question.]) There are 5 computers (hardware). Each computer hosts 4 virtual machines (VM) with different addresses, and each VM hosts 10 applications. How many key exchange sessions (in one session, two logical parties share a key) are needed if the decentralized key distribution is used so that all applications share a pair-wise keys? 3 / 3 points

19900

☒ Correct See Key Hierarchy and Decentralized Key Distribution lecture.

6. Which of the followings are needed for a CA to generate a digital certificate? 2 / 3 points

- ☒ The private key of the user (the certificate subject)
- ☒ This should not be selected See Public-Key Certificate lecture.

- ☐ The private key of CA
- ☒ The request for the user's certificate

☒ Correct See Public-Key Certificate lecture.

- ☒ The public key of the user (the certificate subject)
- ☒ Correct See Public-Key Certificate lecture.

- ☐ The nonce used by the user in the last communication
- ☐ The public key of CA

7. Which of the followings are required to verify the signature of the digital certificate signed by a CA? 3 / 3 points

- ☒ The certificate itself
- ☒ Correct See Public-Key Certificate lecture.

- ☒ The public key of CA
- ☒ Correct See Public-Key Certificate lecture.

- ☐ The nonce involved in the user-CA communication
- ☐ The user's request for the digital certificate
- ☐ The private key of CA

8. Which of the following statements are true? Select all that applies. 4 / 4 points

- ☒ Digital certificates can be requested before using it to share the public key.
- ☒ Correct See Public-Key Certificate lecture.
- ☐ The communication between the Public-Key Authority and the requesting user needs to be secure, so that a third party cannot decrypt and learn the content of the communication
- ☐ Digital certificates need to be accessible only the user that requested for it.
- ☐ Digital certificates remain valid until it explicitly gets revoked.
- ☐ Any user has the authority to revoke a certificate.
- ☒ Once receiving the digital certificates signed by a Certificate Authority (CA), a user can share it with anybody whom it wants to communicate.
- ☒ Correct See Public-Key Certificate lecture.

9. Which of the followings are the PKI's responsibilities? Select all that applies. 3 / 3 points

- ☒ Revoke certificates
- ☒ Correct See Public-Key Certificate and PKI lecture.
- ☒ Distribute certificates
- ☒ Correct See Public-Key Certificate and PKI lecture.
- ☒ Manage certificates
- ☒ Correct See Public-Key Certificate and PKI lecture.
- ☒ Create certificates
- ☒ Correct See Public-Key Certificate and PKI lecture.
- ☒ Store certificates
- ☒ Correct See Public-Key Certificate and PKI lecture.